Cover Letter

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This research was conducted under the guidance of **Professor Mausam** as part of the **Natural Language Processing** course at **IIT Delhi** in 2016. Over a period of three months, we explored methods to enhance plagiarism detection software without relying on prior knowledge of an author's writing style. This work was later made available as a preprint on **viXra** in 2019, and is indexed on **Semantic Scholar**. My contributions to this research included introducing PQ-Gram-based features to capture authors' writing styles, training classifiers over relatively pure and vital segments as a hypothesis rather than relying on the bag-of-words model used in the baseline approach. I implemented and tested this hypothesis, confirming that our proposed model is more robust while maintaining comparable accuracy to the baseline. Additionally, I took the **lead in writing the paper** under the guidance of my co-author, Mr. Sengupta, a prospective Ph.D. student under Professor Mausam. We believe this research provides valuable insights to the NLP and academic integrity communities.

Unsupervised Decomposition of Multi-Author Document: *Exploiting the difference of Syntactic writing styles*

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Abstract

This paper proposes an improvement over a recent *paper*[1]. We have worked on two aspects. In the first aspect, we aim to capture the writing style of an author by employing an n-gram model of words, POS tags, and a PQ-Gram model of syntactic parsing, enhancing the basic unigram model used previously. In the second aspect, we introduce layers of refinement to the existing baseline model and propose a new term, "similarity index," to distinguish between pure and mixed segments before unsupervised labeling. The similarity index uses the overall and sudden changes in writing style, as identified by the PQ-Gram model, and n-gram models applied to lexicalized and unlexicalized sentences within segments for refinement. This paper investigates the role of feature selection in capturing syntactic patterns specific to an author and its overall effect on the final accuracy of the baseline system. Specifically, we insert a refinement layer into the baseline system and define a threshold based on similarity measures between sentences to assess the purity of segments provided as input to the GMM. The key idea of our approach is to provide the GMM clustering algorithm with "good segments" to maximize clustering precision, which is then used as labels to train a classifier. We also experiment with different feature sets, such as bigrams and trigrams of POS tags, and a PQ-Gram-based feature on unlexicalized PCFG to capture distinct writing styles. These features are given as input to a GMM trained by an iterative EM algorithm to generate good clusters of the segments of the merged document.

Introduction

Here, we propose a new unsupervised method for decomposing a multi-author document into authorial components. We assume that no prior information is available about the authors or the document, except for the number of authors. The key idea is to exploit the differences in the grammatical writing styles of the authors and use this information to build paragraph clusters. This is a challenging problem at many levels. It is relatively easy to decompose documents based on topics and contexts, which is often referred to as text segmentation in the literature. However, it becomes difficult to distinguish if multiple authors have written about the same topic. Quantifying the differences in the grammatical writing styles of authors is another significant challenge. Since there is no prior information or access to the authors' written texts, supervised classification approaches cannot be directly applied. Additionally, the number of authors in a given document or article (in the case of plagiarism) is generally unknown, which makes it another major challenge. Therefore, determining the number of clusters is another significant task. Given these constraints, this paper focuses on the feature selection aspect of the text, which is the most critical part of the unsupervised clustering process. Good feature selection leads to more precise clustering, ensuring that correct sentences are assigned to their respective clusters. Traditional studies on text segmentation, as shown in Choi (2000), Brants et al. (2002), Misra et al. (2009), and Henning and Labor (2009), focus on dividing the text into significant components such as words, sentences, and topics, rather than authors. Few approaches, such as those in Schaalje et al. (2013), Segarra et al. (2014), and Layton et al. (2013), deal with documents written by a single author only. Koppel et al. (2011) considered the segmentation of a document according to multiple authors, but this approach requires manual translations and concordance to be available beforehand. Therefore, their method can only be applied to specific types of documents, such as Bible books. Akiva and Koppel (2013) attempted to offer a solution, relying on distance measurement to improve the precision and accuracy of the clustering and classification process. However, their method's performance degrades as the number of authors exceeds two.

Modified Baseline

The latest state-of-the-art technique used is described below: Given a multi-author document written by l authors, it is assumed that each sentence is completely written by only one author. The approach follows these steps:

- Divide the document into segments of fixed length.
- Represent each sentence within a segment as vectors using n-grams of words and PQ grams as the feature set.
- Separate pure and mixed segments by analyzing sudden changes in writing style or words used between sentences

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inside a segment, utilizing the ”similarity index” of the segment.

- Represent the resulting pure segments as vectors using an appropriate feature set (words, POS tags, PQ grams) in the whole merged document, which can differentiate the writing styles among authors.
- Cluster the resulting vectors into l clusters using an appropriate clustering algorithm targeting high recall rates (e.g., GMM with an iterative EM algorithm).
- Re-vectorize the segments using a different feature set to more accurately discriminate between the segments in each cluster.
- Apply the segment elicitation procedure, which identifies the vital segments from each cluster to improve precision rates.
- Re-vectorize all selected segments using another feature set that can capture the differences in writing styles of all the sentences in the document.
- Train the classifier using a Naive Bayes model.
- Classify each sentence using the trained classifier.

Data Set

The data sets we have used to evaluate our model is:

- 690 blogs written by Gary Becker and Richard Posner.
- 1,182 New York Times articles written by Maureen Dowd, Gail Collins, Thomas Freidman and Paul Krugman.
  Each data set has its own set of challenges, since each author has written a lot of different topics and some topics are written by both authors.

The resulting table is shown below:

Table 1: Table Title

Dataset	Accuracy	sentences	Authors
Becker-Posner	0.82	26922	2
GC-TF-PK	0.67	11984	3
MD-TF-PK	0.70	13422	3
MD-GC-PK	0.66	13448	3
MD-GC-TF-PK	0.61	15584	4

Limitations of the Baseline System

We observe that no advanced NLP features are used for this task. The bag-of-words model is a weak approach for effectively discriminating between authors. Additionally, the accuracy of the final stage of classification heavily depends on the chunk (V) of sentences chosen from the individual authors to form the merged document. For instance, changing the parameter (V) from 200 to 50 reduces the final accuracy from 82% to 49%. Training on segments and testing on sentences is not ideal, as the main bottleneck for achieving high accuracy lies in the clustering algorithm. Furthermore,

Figure 1:

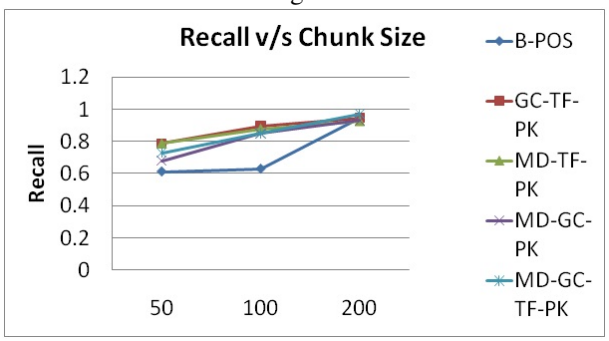
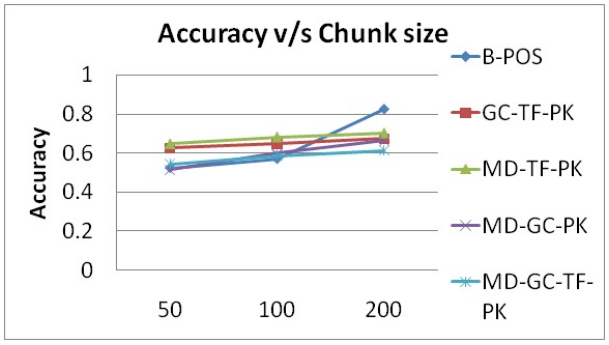


Figure 2:



mixed segments (sentences comprising contributions from both authors) present a challenge during the clustering process, negatively impacting both precision and recall.

Preliminary: PQ Grams

Similar to n-grams, which represent subparts of a given string of length n, p-q grams extract substructures from an ordered, labeled tree. The size of a p-q gram is determined by the stem (p) and base (q). The value of p defines how many nodes are included vertically, while q defines the number of nodes considered horizontally. For example, a valid p-q gram with p=2 and q=3, starting from the PP on the left side of the tree (S2) shown in the Figure 3, would be [PP-NP-DT-JJ-NNS]. The p-q gram index consists of all possible p-q grams within the tree. To obtain all p-q grams, the base is shifted left and right. If fewer than p nodes exist horizontally, the corresponding position in the p-q gram is filled with *, indicating a missing node.

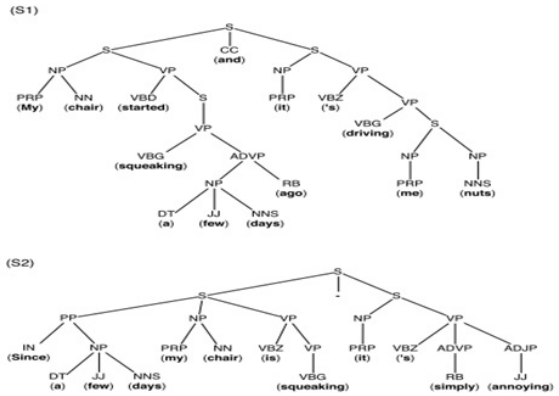
Proposed Methodology

This paper addresses two different aspects. The first aspect focuses on using a PQ-grams-based model, which uses PQ-grams as features of the sentences in the segments and studying its effect on the final accuracy. The second aspect introduces a layer of filtering into the segments to identify which ones are pure (written completely by one author) to the greatest extent possible. This is achieved by defining a threshold value, ideally a similarity score, among the

sentences in each segment. This will help in obtaining better cluster assignments.

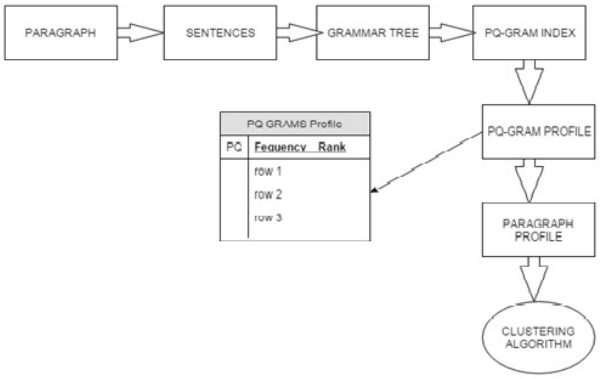
In the first aspect, the main idea is to quantify the differences in grammatical writing styles, which the earlier baseline model lacked, and use this information to build paragraph clusters. By doing this, we aim to decompose sentences effectively. An example below illustrates this. Consider the following two sentences:
S1: My chair started squeaking a few days ago, and it's driving me nuts.
S2: Since a few days ago, my chair has been squeaking - it's simply annoying.

Figure 3: Different parses for similar meaning sentence



The above sentences are semantically similar, although they differ significantly syntactically (as shown in the figure below). A bag-of-words model, which relies on the occurrences of words or word counts, cannot distinguish between these two sentences, as they contain similar types of words. The key idea is to quantify these syntactic differences by calculating grammar profiles and using this information to decompose a collaboratively written document.

Figure 4: The overview of our Method



As shown in the flowchart above, paragraphs are first extracted from the text, and then each sentence is extracted from the paragraphs. For each sentence, a parse tree is formed using the standard StanfordParser. We call this the Grammar tree. From this grammar tree, we extract the PQ Gram indices of these sentences. A p-q gram index of a sentence consists of all possible p-q grams, where multiple occurrences of the same p-q grams are also present multiple times in the index. By combining all p-q gram indices from the sentences, we create a p-q gram profile that contains a list of all p-q grams and their corresponding frequency of appearance in the text. For our experiment, we have used p=2 and q=3. Finally, each paragraph profile is provided as input to a clustering algorithm, which clusters the text based on the p-q grams contained. We also used POS tags from the Penn Treebank as labels. We have not used the head words, as our goal is to capture the structure of the sentence, not the specific choice of words used by each author.
Another feature we explored is the use of POS tags in a bigram and trigram setting to build a paragraph profile, which is then input into a clustering algorithm (GMM-EM).

In the second aspect

- Calculate the similarity between each sentences in a segment by counting the number of common PQ-grams divided by the multiplication of the total PQ grams of each sentences and obtain a score.

- Repeat this process for all sentences in the segment, summing all the scores to obtain a similarity score for the segment. Perform this for all segments.

- This gives us a relative measure of the purity of the segments, where higher segment scores indicate greater purity. In other words, the mixing is biased towards one of the authors.

- With this, we can determine which segments are suitable for the clustering algorithm, leaving the evenly distributed mixed segments with low similarity scores out of the clustering process.

- Train the Gaussian Mixture Model (GMM) on the pure segments, excluding the mixed segments.

- Use the posterior probabilities of the clustered segments to classify the left-out segments.

- Identify vital segments as before and train a Naive Bayes classifier.

- By bypassing some segments from the GMM process, we enhance the selection of vital segments in the baseline system.

Figure 5:

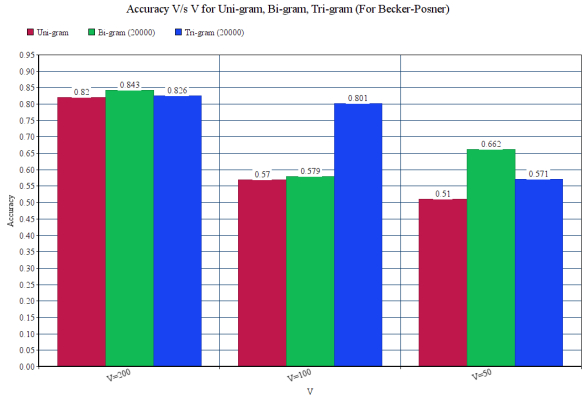
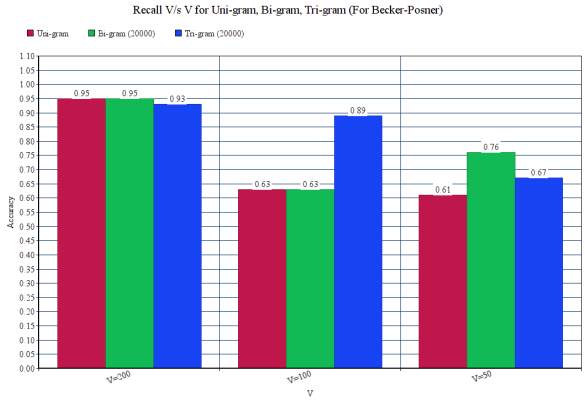


Figure 6:



Results

First Aspect

Table 2: Becker-Posner

Features	Accuracy		
	V=200	V=100	V=50
Baseline	0.82	0.57	0.51
POS-tags(bigram)	0.67	0.64	0.63
POS-tags(trigram)	0.70	0.67	0.65
PQ-GRAMS	0.66	0.63	0.62
PQ-GRAMS+tf-idf	0.69	0.65	0.63

Analysis

When applied to the Becker-Posner dataset (26,922 sentences), we encountered several long sentences that the parser could not process due to memory limitations. As a result, we ignored these sentences (only 3) and evaluated the above strategies on the reduced dataset. Our observations from this experiment suggest that, intuitively, this is a good method for capturing the different syntactic aspects of writers. However, it significantly increases the dimensionality of the feature space compared to the baseline. The baseline methods had a much smaller feature size, making them simpler and faster. Additionally, we found that introducing these features did not necessarily improve the final accuracy of the baseline model.

We observed that the sensitivity of the model with the new features, even after reducing the chunk size (V), remained quite robust. It did not vary drastically, unlike the baseline system. Therefore, we can conclude that these syntactic features are quite stable.

Future Scope

The underlying assumption of this investigation was that the number of authors in the document is known a priori. Future work could focus on determining the number of authors automatically using different clustering algorithms and studying the effect of these features on the discriminating properties of those classifiers.

References

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